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AdventureWorks Business Performance Analysis

Purchasing & Vendor Analytics Perspective | Power BI Summary Report

1. Perspective & Focus Area

This analysis looks at AdventureWorks Cycles from the purchasing side of the business where the company's money goes when it buys materials and components, which vendors it relies on, how reliable those vendors are, and how purchasing connects to shipping costs and product quality. The goal is to give a clear picture of supply-side spending and flag where there might be risk or room to negotiate better.

The model brings together purchase orders, vendor records, product and category data, shipping methods, and vendor lead times, covering total purchasing activity across over 100 active vendors.

2. Key Assumptions

- Completed orders only: Only purchase orders with Status = 4 (Complete) were included, so the figures reflect orders that were actually fulfilled rather than drafts or cancellations.
- Active vendors only: The analysis focuses on vendors marked as active (ActiveFlag = TRUE), since inactive vendors are no longer part of current supply relationships.
- Credit rating labels: AdventureWorks stores vendor credit ratings as numbers (0-4). These were mapped to readable labels, 0 = Superior, 1 = Excellent, 2 = Above Average, 3 = Average, 4 = Below Average - to make the spend-by-risk breakdown easier to read.
- Lead time: Where a vendor supplies multiple products, each with its own lead time, the average lead time across all their products was used as a single per-vendor figure.
- Currency: All monetary values are treated as a single currency without applying currency conversion, as the dataset is predominantly USD-denominated.
- Fulfillment time: Calculated as the difference between order date and ship date on each purchase order, averaged across all orders.
- Order count per vendor: The dataset assigns a uniform count of purchase orders (51) to nearly every vendor, regardless of total spend. This appears to be an artifact of how the AdventureWorks sample data was generated rather than a real purchasing pattern, so "orders per vendor" was not used as a comparative metric in this analysis.

3. Methodology & Derivations

The model connects Purchase Order Header and Purchase Order Detail (the core purchasing records) to Vendor, Ship Method, and Product/Category/Subcategory tables. A separate Product-Vendor table was used to bring in lead time data per vendor. A Date table was added to support year-based filtering and a proper chronological view of monthly spend.

Key calculated measures (DAX):

- Total Purchase Spend = SUM of TotalDue across completed purchase orders - the headline spend figure.
- Total Purchase Orders = DISTINCTCOUNT of PurchaseOrderID - a count of unique completed orders, used per-vendor to compare order volume.
- Top 5 Vendor % of Spend = spend from the five highest-spend vendors divided by total spend a concentration measure.
- Avg Vendor Credit Rating = the average credit rating across active vendors, used alongside the spend-by-rating breakdown.
- Avg Lead Time (Days) and Avg Fulfillment Days = average vendor lead time and average order-to-ship time, used to assess supply chain responsiveness.

- Avg Freight % of Order and Rejection Rate % = freight cost and rejected quantity as a share of order value/received quantity, used to flag cost and quality issues.

4. Insights & Recommendations

Key Insights

1. Most of the money goes to vendors AdventureWorks trusts. About 81% of total purchase spend roughly \$56.9M, goes to vendors rated “Excellent.” That’s reassuring, but the \$6.15M going to “Below Average” vendors is still worth keeping an eye on.
2. Spend is concentrated in a handful of vendors. The top 5 vendors account for about a quarter of all purchasing spend, with names like Superior Bicycles and Chicago City Saddles standing out clearly above the rest, these are the relationships that matter most.
3. Components, especially tires and tubes, dominate the spend. Components as a category make up the bulk of purchase spend, and within that, Tires and Tubes alone account for roughly a third of subcategory spend, this is the category where pricing and supplier terms matter most.
4. Lead times vary a lot, but fulfillment stays steady. Some vendors take well over a month to deliver while others are much faster, yet overall order fulfillment averages just 9 days suggesting AdventureWorks is buffering against slow vendors with inventory rather than relying on speed.
5. Quality and freight costs are both under control. Rejection rates sit around 3% and freight adds only about 2% to order costs overall solid numbers, though a few vendors show noticeably higher rejection rates and are worth a closer look.

Recommendations

- Use the top 5 vendor relationships as leverage. Since a small group of vendors accounts for a large share of spend, these are the best candidates for renegotiating pricing, payment terms, or volume discounts.
- Keep a watch list for lower-rated vendors. The spend going to “Below Average” rated vendors isn’t huge, but it’s worth tracking separately to make sure it doesn’t grow without a corresponding improvement in their rating.
- Prioritize sourcing strategy around Components and Tires/Tubes. Given how much spend sits in this single subcategory, even small efficiency gains here better pricing, alternative suppliers, bulk deals would have an outsized impact.
- Investigate the slowest vendors. A few vendors have lead times far above the rest worth checking whether this is creating hidden inventory costs or risk, even if current fulfillment numbers look fine on average.
- Follow up on vendors with higher rejection rates. A small number of vendors are driving most of the rejected stock, a quality review or corrective action plan with these specific vendors could improve the overall rejection rate further.

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70.48M

Total Purchase Spend

4K

Total Purchase Orders

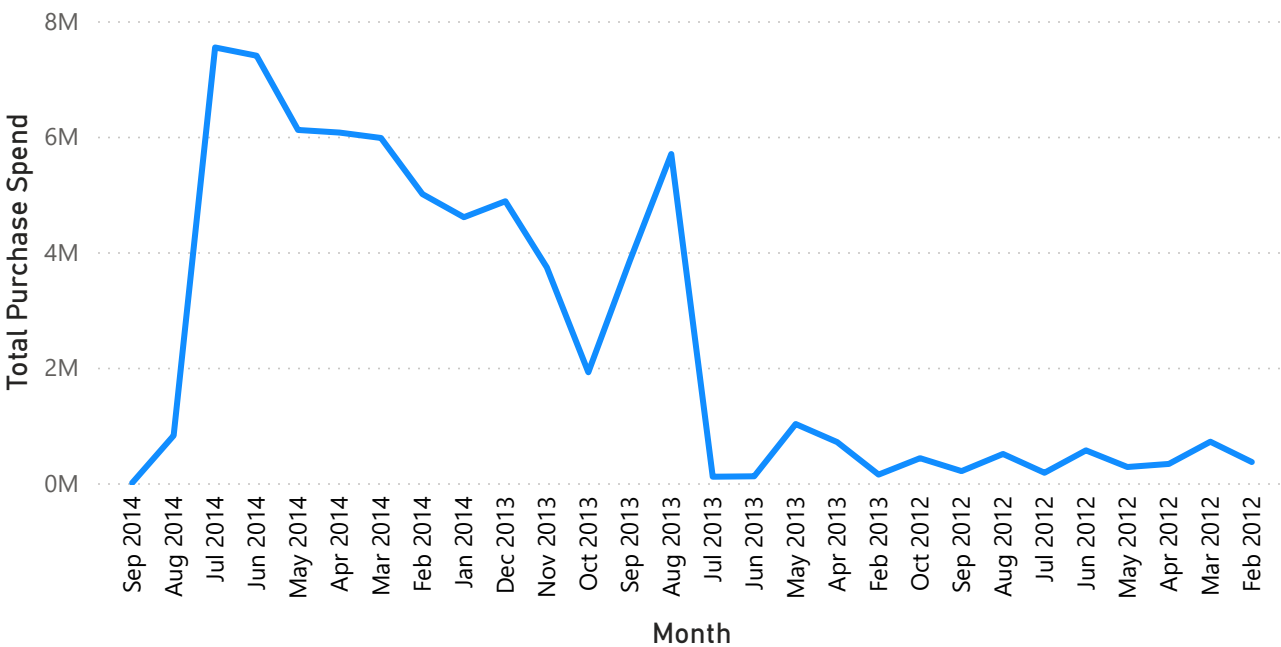
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Total Active Vendors

17.57K

Avg PO Value

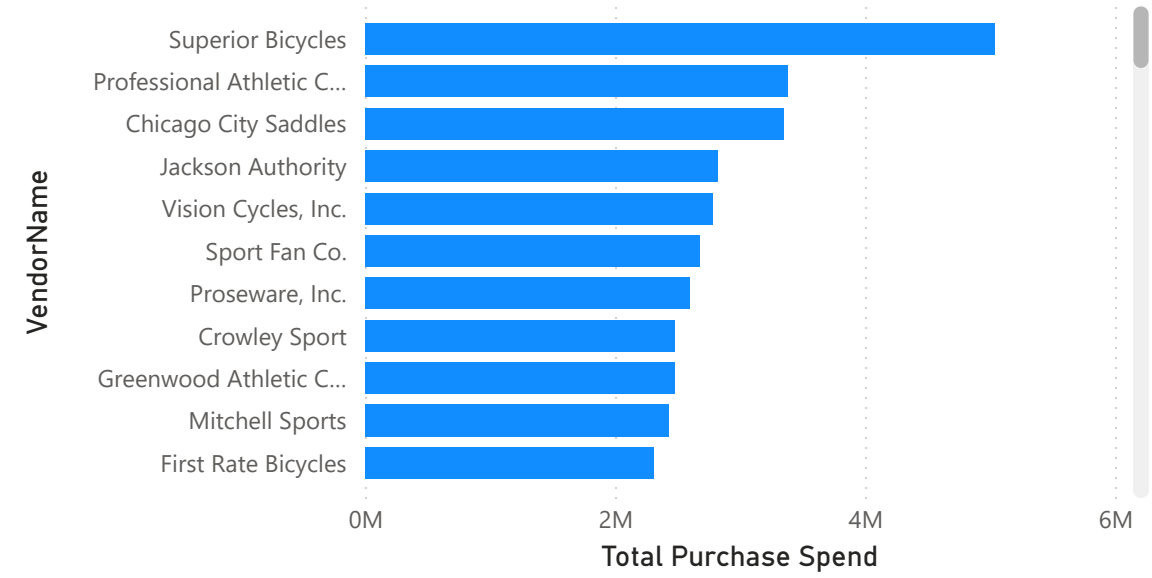
Total Purchase Spend by Month



Year

- ☐ 2011
- ☐ 2012
- ☐ 2013
- ☐ 2014

Total Purchase Spend by VendorName



VendorName	Total Purchase Spend	Credit Rating Label	Total Purchase Orders
Advanced Bicycles	28,502.09	Excellent	51
Allenson Cycles	498,589.59	Above Average	51
American Bicycles and Wheels	9,641.01	Excellent	51
American Bikes	1,149,489.84	Excellent	51
Anderson's Custom Bikes	824,365.17	Excellent	51
Aurora Bike Center	30,229.37	Excellent	51
Australia Bike Retailer	25,060.04	Excellent	51
Beaumont Bikes	79,384.77	Excellent	51
Bergeron Off-Roads	38,622.12	Excellent	51
Bicycle Specialists	1,952,375.30	Excellent	51
Bike Satellite Inc.	87,404.10	Excellent	51
Bloomington Multisport	8,243.95	Excellent	51
Burnett Road Warriors	5,779.99	Excellent	51
Business Equipment Center	8,497.80	Above Average	51
Capital Road Cycles	770,340.44	Excellent	51
Carlson Specialties	561,550.10	Above Average	51
Chicago City Saddles	3,347,165.20	Excellent	51
Chicago Rent-All	31,101.20	Above Average	51
Circuit Cycles	324,301.48	Excellent	51
Comfort Road Bicycles	1,582,013.06	Excellent	51
Compete Enterprises, Inc	1,731,662.68	Excellent	51
Compete, Inc.	350,018.42	Excellent	51
Competition Bike Training Systems	1,594,705.61	Excellent	51
Consumer Cycles	3,378.17	Average	51
Continental Pro Cycles	22,960.07	Average	51
Total	70,479,332.64		4012

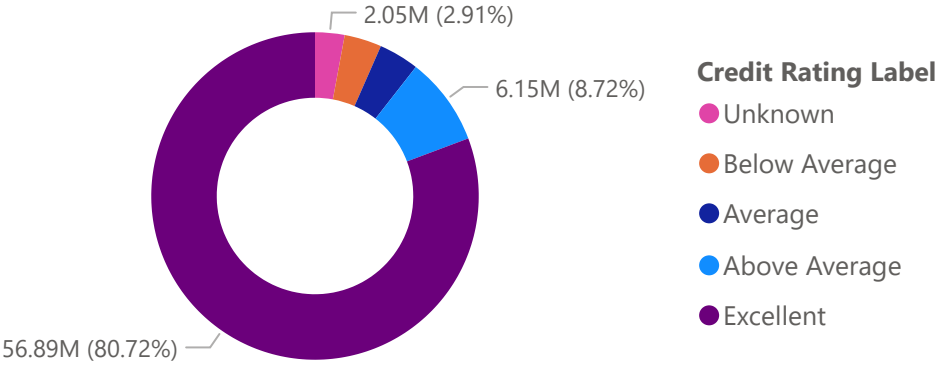
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Top 5 Vendor % of Spend

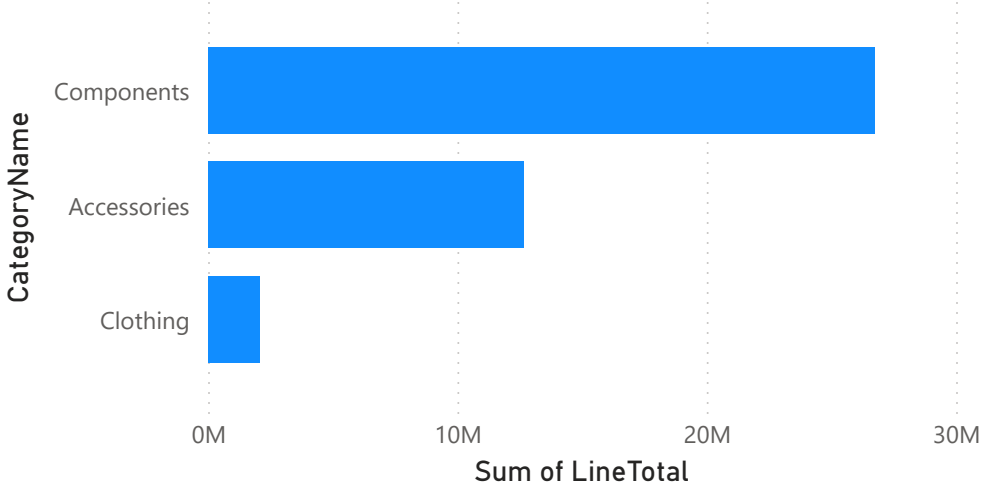
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Avg Vendor Credit Rating

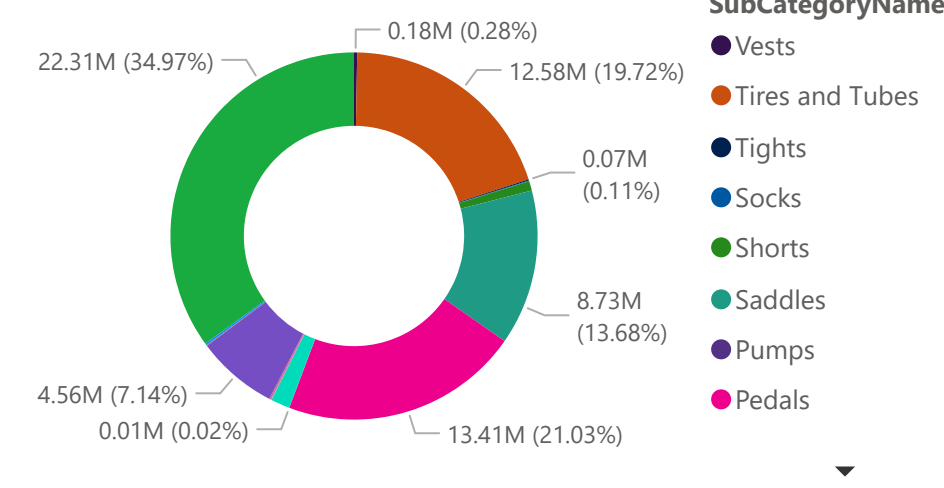
Total Purchase Spend by Credit Rating Label



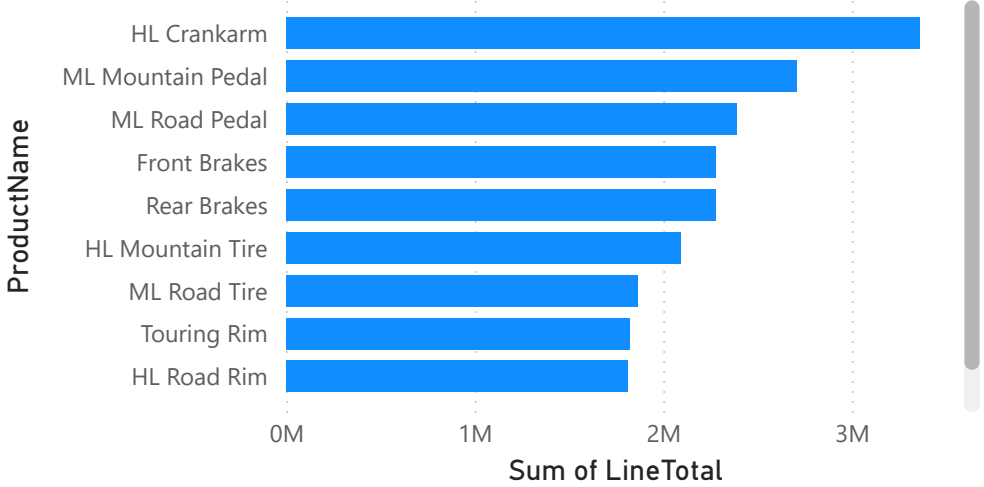
Sum of LineTotal by CategoryName



Sum of LineTotal by SubCategoryName



Sum of LineTotal by ProductName



Year

- ☐ 2011
- ☐ 2012
- ☐ 2013
- ☐ 2014

19.45

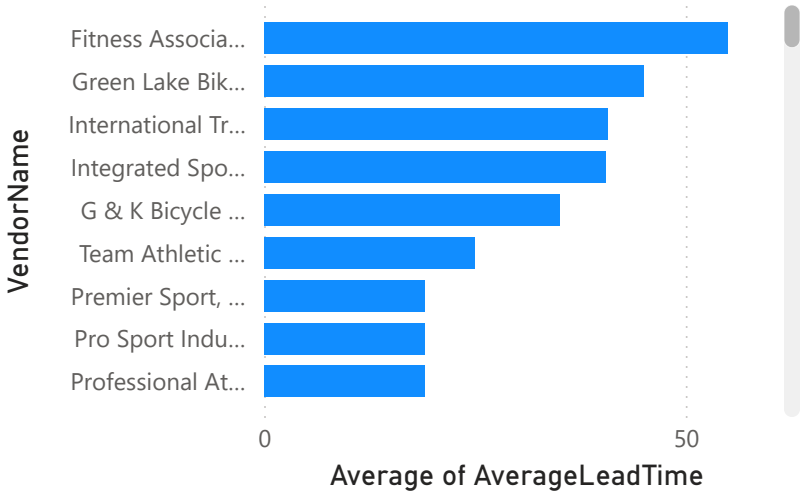
Avg Lead Time (Days)

9.05

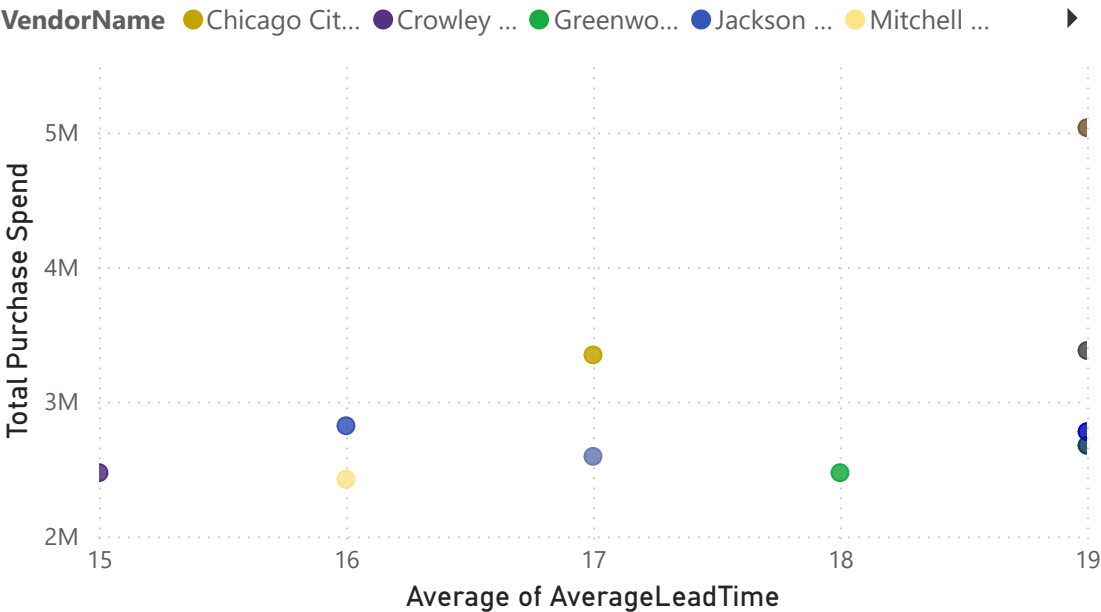
Avg Fulfillment Days

VendorName	AverageLeadTime	Total Purchase Spend
Superior Bicycles	10	5,034,266.74
Superior Bicycles	12	5,034,266.74
Superior Bicycles	15	5,034,266.74
Superior Bicycles	16	5,034,266.74
Superior Bicycles	17	5,034,266.74
Superior Bicycles	18	5,034,266.74
Superior Bicycles	19	5,034,266.74
Superior Bicycles	25	5,034,266.74
Superior Bicycles	30	5,034,266.74
Superior Bicycles	35	5,034,266.74
Superior Bicycles	45	5,034,266.74
Total		70,479,332.64

Average of AverageLeadTime by VendorName



Average of AverageLeadTime and Total Purchase Spend by VendorName



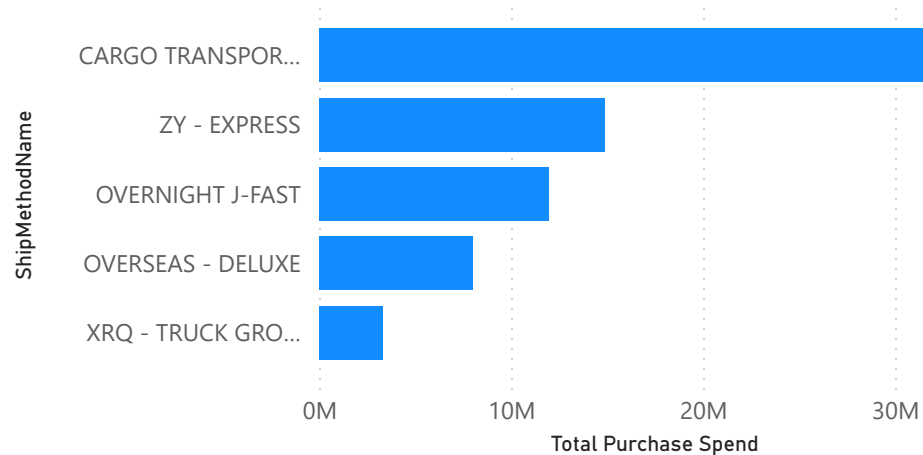
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Avg Freight % of Order

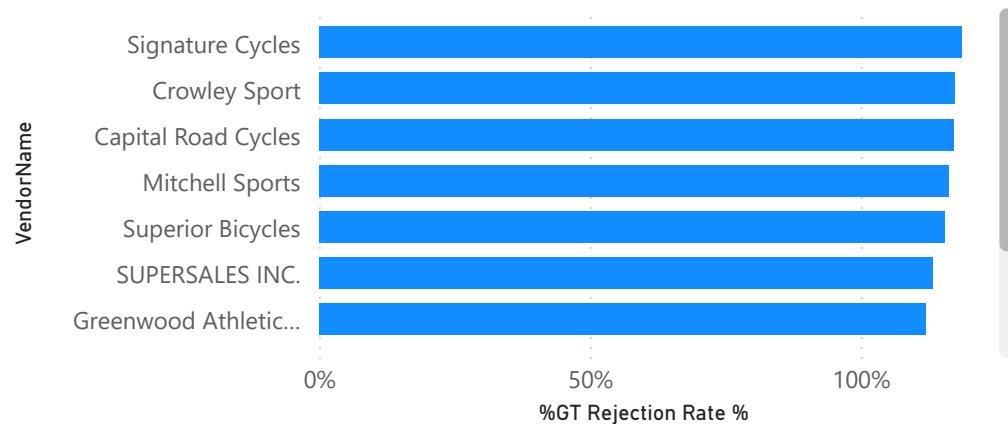
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Rejection Rate %

Total Purchase Spend by ShipMethodName



%GT Rejection Rate % by VendorName



1. 81% of spend goes to "Excellent"-rated vendors supply base is largely low-risk.
2. Top 5 vendors = ~25% of total spend is on Superior Bicycles and Chicago City Saddles lead.
3. Components, especially tires/tubes, dominate spend nearly a third of subcategory total.
4. Lead times vary widely, but fulfillment stays at 9 days buffers absorb vendor delays.
5. Rejection rate (3%) and freight cost (2%) are both low, a few vendors stand out for higher rejections.